Set-up: L layers with layer ℓ having weights $W^{(\ell)} \in M_{m_\ell \times m_\ell}(\mathbb{R})$, biases $b^{(\ell)} \in \mathbb{R}^{m_\ell}$, and activation $\sigma_\ell : \mathbb{R} \to \mathbb{R}$ defined component-wise when applied to vectors.

Loss function: f-n of all $W_{ij}^{(\ell)}$ and $b_i^{(\ell)}$

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^{T} \| \hat{y}^{(t)} - y^{(t)} \|^2$$

for simplicity start with $T=1$. Can generalize later. Then

$$\mathcal{L} = \| \hat{y} - y \|^2 = \sum_{c=1}^{m_L} (\hat{y}_c - y_c)^2 = \sum_{c=1}^{m_L} (a_c^{(L)} - y_c)^2 \qquad *(1)$$

Now we have $\mathcal{L}$ in terms of $a_c^{(L)}$'s for $c = 1, \dots, m_L$. Now with that in mind, and looking at the graph of the NN:

$$\underset{\text{apply weights}}{a^{(\ell-1)} \Rightarrow z^{(\ell)} := W^{(\ell)} a^{(\ell-1)} + b^{(\ell)}} \Rightarrow \underset{\text{apply activation}}{a^{(\ell)} := \sigma_\ell (z^{(\ell)})}$$

For $p = W_{ij}^{(\ell)}$ or $b_i^{(\ell)}$ with j free and i fixed,

$$\frac{\partial \mathcal{L}}{\partial p} = \sum_{c=1}^{m_\ell} \frac{\partial \mathcal{L}}{\partial a_c^{(\ell)}} \cdot \frac{\partial a_c^{(\ell)}}{\partial p} = \sum_{c=1}^{m_\ell} \frac{\partial \mathcal{L}}{\partial a_c^{(\ell)}} \frac{\partial a_c^{(\ell)}}{\partial z_c^{(\ell)}} \cdot \frac{\partial z_c^{(\ell)}}{\partial p}$$

Now note that $z_c^{(\ell)} = b_c^{(\ell)} + W_{c1}^{(\ell)} a_1^{(\ell-1)} + W_{c2}^{(\ell)} a_2^{(\ell-1)} + \dots + W_{c m_\ell} a_{m_\ell}^{(\ell-1)}$

So if $c \neq i$, then $\frac{\partial z^{(\ell)}}{\partial p} = 0$, thus

$$\frac{\partial \mathcal{L}}{\partial p} = \frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}} \cdot \frac{\partial a_i^{(\ell)}}{\partial z_i^{(\ell)}} \cdot \frac{\partial z_i^{(\ell)}}{\partial p}$$

Now if $p = W_{ij}^{(\ell)}$ then $\frac{\partial z_i^{(\ell)}}{\partial p} = a_j^{(\ell-1)}$

if $p = b_j^{(\ell)}$ then $\frac{\partial z_i^{(\ell)}}{\partial p} = 1$

we also have $a_i^{(\ell)} = \sigma_\ell(z_i^{(\ell)})$ so $\frac{\partial a_i^{(\ell)}}{\partial z_i^{(\ell)}} = \sigma_\ell'(z_i^{(\ell)})$

Thus:

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}} \cdot \sigma_\ell'(z_i^{(\ell)}) \cdot a_j^{(\ell-1)} \qquad \circledast$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}} \cdot \sigma_\ell'(z_i^{(\ell)}) \qquad \circledast$$

Now we want $\frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}}$. First suppose $\ell = L$; we looking at last layer

Recall from $*(1)$, with only one training example

$$\mathcal{L} = \sum_{c=1}^{m_L} (a_c^{(L)} - y_c)^2$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial a_i} = 2(a_i^{(L)} - y_i) \qquad \circledast$$

Now for $\ell < L$. Note $\ell+1$ is closer to L than ℓ, thus all components of $a^{(\ell+1)}$ depend on $a_i^{(\ell)}$. Thus:

$$\frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}} = \sum_{k=1}^{m_{\ell+1}} \frac{\partial \mathcal{L}}{\partial a_k^{(\ell+1)}} \cdot \frac{\partial a_k^{(\ell+1)}}{\partial a_i} = \sum_{k=1}^{m_{\ell+1}} \frac{\partial \mathcal{L}}{\partial a_k^{(\ell+1)}} \cdot \frac{\partial a_k^{(\ell+1)}}{\partial z_k^{(k+1)}} \cdot \frac{\partial z_k^{(k+1)}}{\partial a_i^{(\ell)}}$$

As above, $\frac{\partial a_k^{(\ell+1)}}{\partial z_k^{(\ell+1)}} = \sigma_{\ell+1}'(z_k^{(\ell+1)})$ and $\frac{\partial z_k^{(\ell+1)}}{\partial a_i^{(\ell)}} = \frac{\partial}{\partial a_i^{(\ell)}}(b_k^{(\ell+1)} + w_{k1}^{(\ell+1)} a_k^{(\ell)} + \cdots)$

$$= w_{ki}^{(\ell+1)}$$

And so

$$\frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}} = \sum_{k=1}^{m_{\ell+1}} \frac{\partial \mathcal{L}}{\partial a_k^{(\ell+1)}} \cdot \sigma_{\ell+1}'(z_k^{(\ell+1)}) \cdot w_{ki}^{(\ell+1)} \qquad \circledast$$

So we can calculate $\frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}}$ recursively.

Now what if we had more than one training example?

$$\therefore \mathcal{L} = \frac{1}{T} \sum_{t=1}^{T} \| \hat{y}^{(t)} - y^{(t)} \|^2 = \frac{1}{T} \sum_{t=1}^{T} \sum_{c=1}^{M_L} (\hat{y}_c^{(t)} - y_c^{(t)})^2$$

$$= \frac{1}{T} \sum_{t=1}^{T} \sum_{c=1}^{M_L} (a_c^{(L)(t)} - y_c^{(t)})^2$$

Now there's multiple $a^{(L)}$'s, indexed by t, so we need to reflect that in our chain rule:

$$\frac{\partial \mathcal{L}}{\partial p} = \frac{1}{T} \sum_{t=1}^{T} \sum_{c=1}^{M_\ell} \frac{\partial \mathcal{L}}{\partial a_c^{(\ell)(t)}} \frac{\partial a_c^{(\ell)(t)}}{\partial z_c^{(\ell)(t)}} \frac{\partial z^{(\ell)(t)}}{\partial p}$$

$$= \cdots$$

$$= \frac{1}{T} \sum_{t=1}^{T} \frac{\partial \mathcal{L}^*(t)}{\partial p}$$

where we define $\mathcal{L}^*(t)$ to be the loss function on one example with training data $\# t$

Since there is no dependency between $a^{(\ell_1)(t_1)}$ and $a^{(\ell_2)(t_2)}$ for $t_1 \neq t_2$, then our formulas for $\frac{\partial \mathcal{L}}{\partial a_i^{(\ell)}}$ stay the same. We just have to calculate lots of them...